ER. ROSHANI GIRI

Backend Developer | Python & Data Science
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PROFESSIONAL SUMMARY

Backend Developer (BE in Information Technology) with hands-on experience building and maintaining production REST APIs using Node.js, Express.js, MySQL and Firebase. Delivered backends for an online examination platform and a hospital management system, covering authentication, authorization, validation, error handling and third-party integrations. Works closely with frontend teams on API contracts, debugs and resolves reported issues, and writes clear, maintainable code. Also skilled in Python and data science, with training and projects in machine learning and computer vision.

SKILLS

Backend & APIs: Node.js, Express.js, Next.js, RESTful API Design, JavaScript (ES6+), Firebase Functions, JWT, Nodemailer

Databases & Storage: MySQL, PostgreSQL, Firebase Realtime Database, SQL/NoSQL Schema Design, Query Optimization, Cloudinary

Secure Development: Authentication, Authorization, Input Validation, Error Handling, Third-Party API Integration

Tools & Collaboration: Git, GitHub, Docker, Postman, Jira, VS Code, Jupyter Notebook, Figma (design handoff), Frontend/Backend Integration

Additional: Python, Pandas, NumPy, scikit-learn, Matplotlib, Seaborn, OpenCV, SIFT, DBSCAN

PROFESSIONAL EXPERIENCE

Backend Developer | Blackmoon Technology, Lalitpur, Nepal May 2025 – Present

- Develop and maintain RESTful APIs using Node.js and Express.js for production application workflows.
- Led backend implementation of the Agnikunda online examination platform, including business logic, authentication, authorization and data flows.
- Built Hospital Management System features covering patient records, doctor management, appointments and related data processing.
- Designed and optimized SQL and NoSQL database schemas and queries to meet application requirements.
- Applied validation and error-handling practices to improve API reliability and consistency.
- Integrated third-party services and file storage using Cloudinary and Firebase.
- Collaborated with frontend developers to define API behavior, integrate components and resolve integration issues.

PROJECTS

Agnikunda – Online Examination Platform Client Project
Technologies: *Node.js, Express.js, MySQL, REST APIs, JWT, Nodemailer*

- Developed backend APIs for student registration, authentication, instructor management, question organization and quiz performance history.
- Implemented subscription-based content access, expiry checks, purchase history, repurchase validation and OTP-based password reset.

Hospital Management System Professional Project
Technologies: *Node.js, Express.js, JavaScript, MySQL, REST APIs*

- Developed and maintained APIs for patient registration, OPD/IPD visits, consultations, admissions, billing, pharmacy, laboratory services and medical records.
- Implemented validation and database operations to keep data consistent across hospital workflows; investigated and resolved reported issues.

Deadstock – MERN Stack E-Commerce Application Academic Project
Technologies: *Express.js, React.js, Node.js, Firebase Authentication, Cloudinary*

- Built product listing, cart management and order workflows for a full-stack e-commerce application.
- Implemented authentication, product CRUD APIs, user management, order processing and product image storage.

Sahaj Yatri – Travel Management Platform
Technologies: *Node.js, Express.js, MySQL, REST APIs*

- Contributed to backend features, REST API integration, database operations and application business logic.

Copy-Move Image Forgery Detection (CMFD) Academic Project
Technologies: *Python, OpenCV, SIFT, DBSCAN, NumPy*
GitHub: github.com/Giri-Roshni/cmfd_miniproject

- Built a SIFT-based keypoint detection workflow and used DBSCAN clustering to identify duplicated image regions.
- Generated forgery masks and annotated images to highlight suspected tampering.

Image Classification Web App (MobileNetV2 + CIFAR-10)
Technologies: *Python, TensorFlow, MobileNetV2, Streamlit, Pillow*
GitHub: github.com/Giri-Roshni/image_classification

- Built a Streamlit web app where users upload JPG, JPEG or PNG images and receive the predicted class with a confidence score, using a MobileNetV2 model fine-tuned on CIFAR-10.
- Preprocessed images to match MobileNetV2 input requirements; achieved about 88–90% accuracy on visually similar classes (cat/dog, deer/horse).

EDUCATION

Bachelor of Engineering in Information Technology (BE IT) 2022 – 2026
Pokhara University, Everest Engineering College, Lalitpur, Nepal

TRAINING

Data Science Training 3 Months
Data cleaning, preprocessing, exploratory data analysis, visualization, supervised machine learning, feature engineering and computer vision using Python, Pandas, NumPy, scikit-learn, Matplotlib, Seaborn and OpenCV.
Certificate: mindrisers.com.np/certificate/validate/MR-82015-DS

LANGUAGES

English (Professional) | Nepali (Native) | Hindi (Conversational)